

Hridik Hingorani

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

B.S. Computer Engineering, Minor: Mathematics

Aug. 2023 – May 2027

- **Coursework:** Robotics (ECE 470), Control Systems (ECE 486), Principles of Safe Autonomy (ECE 484), Operating Systems (ECE 391), Deep Learning for Computer Vision (ECE 494)

EXPERIENCE

Software Associates

Remote

AI Software Engineering Intern

May 2026 – Aug. 2026

- Built a local-first voice assistant on embedded Linux and Apple Silicon: Whisper STT, LLM routing through Ollama with cloud fallback, Coqui TTS, FastAPI service layer, and an Electron client; ran MLX LoRA fine-tuning to adapt the local model's response behavior.
- Diagnosed and fixed audio-pipeline timing failures, device interface faults, and I2C/UART peripheral errors that were dropping utterances mid-stream, taking capture from intermittent to reliable end-to-end.
- Shipped a Python/Flask trading service on the Interactive Brokers REST API: five-indicator weighted signal engine (RSI, MACD, Bollinger Bands, moving averages, ATR), automated stop-loss / take-profit / trailing-stop risk logic, and a live position and P&L dashboard.

University of Illinois – ECE Department

Urbana, IL

Course Assistant, ECE 385 (FPGA Design)

Jan. 2026 – Present

- Lead lab sections for 60+ students on SystemVerilog, FPGA timing closure, and hardware-software co-design; debug student implementations across RTL logic, timing constraints, and board-level signal integrity.

PROJECTS

Flex-Sensor Teleoperated Hand | *ESP32, C++, KiCad, ESP-NOW*

July 2026 – Present

- Designed and fully routed a 5-channel flex-sensor interface board in KiCad: ESP32 restricted to ADC1-only pins (GPIO32–36) to avoid ADC2/WiFi contention, 50 k Ω dividers per channel, and a PCA9685 PWM driver for servo output.
- Wrote the firmware stack (ADC sampling, ESP-NOW leader-follower link, PWM servo mapping) and breadboard-validated the full glove-to-hand signal path, catching hardware bugs before committing the board to fabrication.

Real-Time Task-Space Impedance Control – CRS Arm | *C, TMS320F28335 DSP*

Spring 2026

- Implemented a **1 kHz** task-space impedance controller in C issuing joint torque commands to a 3-joint arm: task-space PD law mapped through $A = J^T R$, per-waypoint stiffness modes for selective compliance, Coulomb and viscous friction compensation, and ± 5 N·m torque saturation.
- Estimated joint velocity by finite-differencing encoder angles through a 3-sample moving average; applied open-loop feedforward normal force via estimated contact stiffness with no force sensor, verified against a physical scale. Debugged actuator saturation, loop-timing violations, and near-singular configurations on an oscilloscope.

RISC-V Preemptive Multitasking OS Kernel | *C, RISC-V Assembly*

Spring 2026

- Built a ~6,000-line bare-metal kernel from scratch: preemptive scheduler with deterministic context switching, Sv39 three-level virtual memory, VirtIO block device driver, FAT filesystem, fork/exec/exit, pipes, and an interactive shell.
- Traced a page-table entry truncation bug through GDB and QEMU that silently corrupted physical page numbers on high-address mappings; isolated it to PTE field packing and validated the fix across the full paging path.

UR3 Vision-Guided Contour Drawing | *Python, ROS, OpenCV*

Spring 2026

- Derived closed-form analytic inverse kinematics for a 6-DOF UR3 from scratch alongside product-of-exponentials forward kinematics via matrix exponentials; commanded joint targets over ROS held to 0.0005 rad per joint.
- Built the vision front end (Canny edge detection, contour extraction, polygonal simplification) mapping an arbitrary image onto a 20 cm workspace square; executed a 142-stroke test drawing in 7 minutes on physical hardware.

Cart-Pole Control Simulator | *Rust*

Summer 2026

- Wrote an LQR solver from scratch (numerical linearization, discrete Riccati iteration) with no external math libraries, then benchmarked 6 gain presets against tuned PID on settling time and actuator effort over full nonlinear dynamics integrated at 1 kHz with RK4.

TECHNICAL SKILLS

Robotics & Controls: closed-form and analytic IK, product of exponentials, impedance and compliance control, LQR, PID, RK4 integration, motion planning (A*, RRT*), ZMP gait planning, ROS, URDF

Languages: C/C++, Python, SystemVerilog, RISC-V Assembly, MATLAB, Rust (application-level)

Embedded & Hardware: real-time Linux, bare-metal firmware, preemptive scheduling, I2C/UART/ADC/GPIO/PWM, SPI (loopback), ESP32, Raspberry Pi, FPGA, KiCad, Git